Combined foot patrol of track and OHE

Target features and user journey map. One patrol walks the beat once, inspecting the track below and the 25 kV OHE above, and each finding is routed to its own department. This map shows what the Beta must let the patroller, TPC and supervisors do, step by step, and which of the 19 target features carries each step. OHE content follows RDSO TI/MI/0060 (para 6.1) and the CAMTECH OHE handbooks. Screens are rendered from the app's Compose code in CI.

4

journeys: combined patrol, track danger, OHE danger, routed review and closure

19

target features, each with an acceptance test

93

defect classes, bilingual: 67 track and bridge, 26 OHE

2

danger paths: protect the line (14 track classes) or treat as live and report to TPC (2 OHE classes)

01 · Core workflow

Thirteen stages, who acts, what each stage hands on, both danger branches and the routing to P.Way and TRD.

02–05 · Journey maps

J1 combined patrol, J2 track danger, J3 OHE danger, J4 routed review and closure.

06–07 · Target features

F01–F19 grouped by capability, with implementation, acceptance test and journey coverage.

One walk, two departments, one audited record

Each stage hands a defined deliverable to the next. While walking, the patroller adds at most one tap or one spoken word per finding; classification happens only once stopped. At sync every finding is routed by its class: track to SSE/P.Way, OHE to SSE/TRD (OHE depot). Dashed stages need no user effort.

01	02	03	04	05	06	07	08	09	10	11	12	13
Plan beat	**Kit check**	**Start patrol**	**Walk: track + OHE**	**Capture**	**Quality gate**	**Classify & locate**	**Save offline**	**Sync & route**	**Review & severity**	**Task & repair**	**Close with proof**	**Audit & history**
Patrol	Patrol	Patrol	Patrol	Patrol	App	Patrol	App	App	SSE/JE P.Way · TRD	SSE/JE · gang / depot	SSE/JE	System · ADEN / ADEE
L1 context	Kit record	Patrol record	Patrol trace	L1 evidence	Pass / retake	L2 finding	Sync queued	To P.Way / TRD	L3 decision	L4 task	L4 closure	L5 history

FIELD · COMBINED PATROLLER · OFFLINE ON THE PHONE | NETWORK | OFFICE · SSE/JE P.WAY AND TRD · BACKEND

J1 Combined patrol · stages 01–08

J2 Track danger · protect first, then 05–08

J3 OHE danger · TPC first, then 05–08

J4 Routed review and closure · 09–13

TRACK DANGER (BETWEEN 04 AND 05)

Rail or weld fracture, washaway, obstruction and other protect-first classes: capture stays locked until the patroller confirms **line protected** and **Station Master informed**. Both are timestamped (rule R2).

OHE DANGER (BETWEEN 04 AND 05)

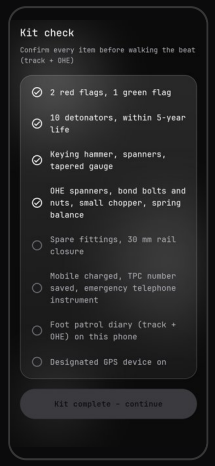
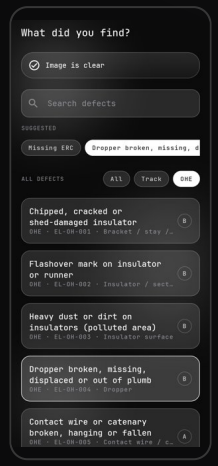
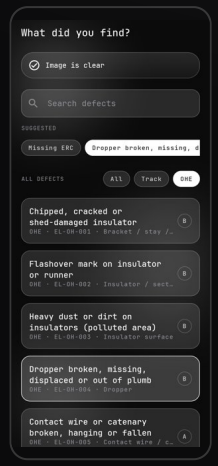
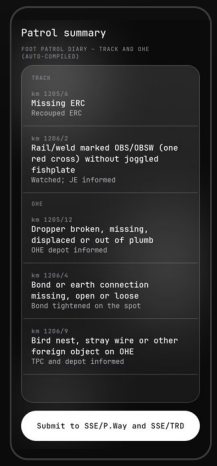
Contact wire or catenary broken or hanging, or a branch or object fouling the pantograph path: **treat as live**. Call TPC, then confirm **TPC informed**, keeping clear and SM informed (RDSO TI/MI/0060 6.1(5); rule R2).

RULES THE BACKEND ENFORCES

R1 Beta classes only
R2 Danger first
R3 AI = candidates
R4 Only SSE/JE decide
R5 Closure needs proof
R6 Hashed audit

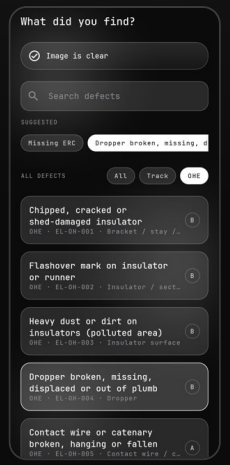
Combined foot patrol: track and OHE (CP-01)

Walk the beat once, look down at the track and up at the OHE, record every defect with photo and location, and hand a diary in two parts to SSE/P.Way and SSE/TRD.

STEP	01 Open app	02 Kit check	03 Walk the beat	04 Missing ERC (track)	05 Broken dropper (OHE)	06 Record action	07 Save	08 End patrol
SCREEN								
USER DOES	Sees today's beat, patrol type CP-01, sync state	Ticks eight items: track kit, OHE tools, phone with TPC number	Scans track, OHE and approaching trains; app stays silent	Captures without stopping	Zooms up, captures, names it	Notes what was done (ERC recouped, bond tightened)	Hears and feels the confirmation, walks on	Reviews the diary: track part and OHE part
INTERACTION	Tap Start patrol · 'start patrol' / 'गश्त शुरू'	Continue unlocks only when all ticked	Live camera behind glass HUD; km in large figures	Tap Capture · 'capture' / 'फोटो लो'	'dropper broken' / 'ड्रॉपर टूटा'; OHE filter; mast number	Optional dictated note · 'note …' / 'टिप्पणी …'	Haptic + spoken 'saved'; back to HUD	One Submit · 'end patrol' / 'गश्त समाप्त'
DELIVERABLE	–	Kit record	Patrol trace	L1 image + GNSS	L2 finding EL-OH-004	Diary entry	Sync queued	Diary routed to P.Way and TRD
FEATURES	F10 F16 Offline store and sync Language switch	F09 Kit check	F08 F17 Location Accessibility	F01 F02 F04 Evidence capture Image-quality gate Voice commands	F01 F03 F04 F08 Evidence capture Bilingual defect taxonomy Voice commands	F05 F06 Voice dictation of notes Spoken and haptic confirmation	F06 F10 Spoken and haptic confirmation Offline store and sync	F11 F10 Combined patrol diary Offline store and sync

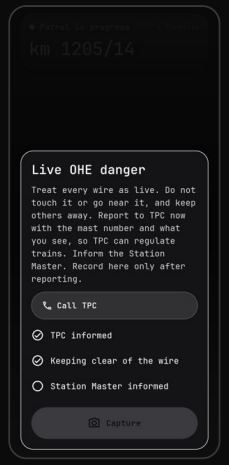
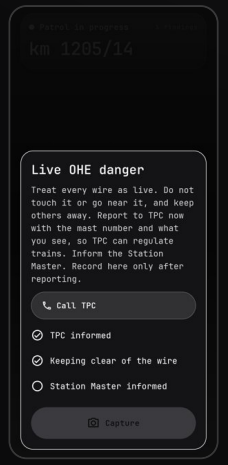
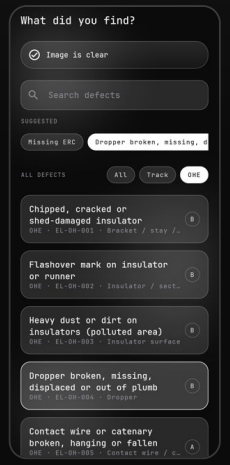
Track danger: rail fracture, washaway

Protect the line first, inform the Station Master, then record evidence with protection times.

STEP	01 Notice danger	02 Protect the line	03 Confirm protection	04 Capture evidence
SCREEN				
USER DOES	Stops; chooses Track	Follows on-screen protection steps	Ticks 'line protected' and 'SM informed'	Photographs the defect after protecting
INTERACTION	Tap Danger · 'danger' / 'खतरा' (opens the chooser only)	Opaque sheet: flags, detonators 600 m and 1200 m (BG), inform SM	Deliberate taps only - no voice path	Capture unlocks after both ticks; class marked protect-first
DELIVERABLE	–	–	Escalation record + timestamps	L1/L2 with protection times
FEATURES	F04 F07 Voice commands Danger guard (track and OHE)	F07 Danger guard (track and OHE)	F07 F14 Danger guard (track and OHE) Audit trail	F01 F02 F08 F10 Evidence capture Image-quality gate Location Offline store and sync

OHE danger: wire down, branch on the wire

Treat the OHE as live, keep clear, report to TPC so trains can be regulated, then record evidence from a safe distance.

STEP	01 Notice danger	02 Keep clear, call TPC	03 Confirm	04 Capture evidence
SCREEN				
USER DOES	Stops well clear; chooses OHE	Does not approach; calls TPC with km, mast, what is seen	Ticks 'TPC informed', 'keeping clear', 'SM informed'	Zooms from a safe distance, then classifies
INTERACTION	Tap Danger · 'danger' / 'खतरा' (opens the chooser only)	Call TPC opens the dialler (number from Settings)	Deliberate taps only – no voice path	Capture unlocks after the ticks; class marked TPC first
DELIVERABLE	–	**TPC call**	**Escalation record + TPC time**	**L1/L2 with TPC time**
FEATURES	F04 F07 Voice commands Danger guard (track and OHE)	F19 F07 Call TPC Danger guard (track and OHE)	F07 F14 Danger guard (track and OHE) Audit trail	F01 F02 F08 F10 Evidence capture Image-quality gate Location Offline store and sync

Routed review and closure

Each department sees only its findings, decides severity from evidence without a site visit, gets the repair done and closes with proof.

STEP	01 Open inbox	02 Assess	03 Create task	04 Close
SCREEN				
USER DOES	Sees findings routed to the department, with photo	Assigns severity (OHE Category A or B); confirms or rejects	Raises work order for gang or OHE depot	Closes only with closure photo
INTERACTION	'To SSE/TRD' or 'To SSE/P.Way'; duplicates grouped	Confirm only after severity is set; AI candidates hidden (shadow)	Work-order reference; TMS / depot register export	Closure image must pass the quality gate (rule R5)
DELIVERABLE	–	L3 decision	L4 task	L4 closure + audit
FEATURES	F12 F18 Routed supervisor review Duplicate suppression	F12 F15 Routed supervisor review AI candidates (shadow)	F12 Routed supervisor review	F13 F02 F14 Closure with evidence Image-quality gate Audit trail

Nineteen features, each with a test that proves it

Filled dots show the journeys a feature serves. "Shadow" means the AI runs and stores candidates but users do not see them until field thresholds are met; "Server" runs in the backend only.

ID	FEATURE	IMPLEMENTATION (STANDARD LIBRARY FIRST)	ACCEPTANCE	TRACES TO	J1	J2	J3	J4	STATUS
CAPTURE & EVIDENCE									
F01	**Evidence capture**	CameraX; JPEG to app storage; SHA-256 on upload	Capture ≤ 1 s from tap; works offline	REQ-03, REQ-14	●	●	●	○	Beta
F02	**Image-quality gate**	Device + server, same metrics; 'insufficient evidence' outcome	Thresholds in Ch. 13	REQ-06	●	●	●	●	Beta
F15	**AI candidates (shadow)**	ONNX detector behind quality gate; candidates stored, not shown until thresholds met	Feature flag	REQ-05	○	○	○	●	Shadow
CLASSIFY & LOCATE									
F03	**Bilingual defect taxonomy**	93 Beta classes (67 track/bridge, 26 OHE), Track/OHE filter, search in both scripts, danger and measurement tags	Every class has Hindi label; ≥ 20 OHE classes (unit tests)	D02; RDSO TI/MI/0060 6.1	●	○	○	○	Beta
F08	**Location**	Fused Location + km or OHE mast number entry + OCR assist (ML Kit Devanagari)	Inspector confirms	REQ-03	●	●	●	○	Beta
F18	**Duplicate suppression**	GNSS ≤ 25 m + embedding similarity (server)	Benchmark Ch. 11	—	○	○	○	●	Server
VOICE & FEEDBACK									
F04	**Voice commands**	On-device SpeechRecognizer hi-IN / en-IN + closed vocabulary matcher	Intent accuracy thresholds (Ch. 13)	REQ-10	●	●	●	○	Beta
F05	**Voice dictation of notes**	Same recogniser; note language stored	Note saved with language tag	PR-07	●	○	○	○	Beta
F06	**Spoken and haptic confirmation**	TextToSpeech + HapticFeedback CONFIRM	Toggle in Settings	REQ-10	●	○	○	○	Beta

Record, review and access

Thresholds referenced as Ch. 13 and benchmark results as Ch. 11 are in the Solution Design Report.

ID	FEATURE	IMPLEMENTATION (STANDARD LIBRARY FIRST)	ACCEPTANCE	TRACES TO	J1	J2	J3	J4	STATUS
SAFETY									
F07	**Danger guard (track and OHE)**	Danger chooser; track: protect line + SM; OHE: treat as live, Call TPC, keep clear, SM; API rule R2; no voice path	Unit + API tests	REQ-01; RDSO TI/MI/00606.1(5)	○	●	●	○	Beta
F19	**Call TPC**	Dialler opens with the TPC number from Settings; the call itself stays a phone call	TPC time recorded before an OHE danger can be saved	RDSO TI/MI/00606.1(5),(7)	○	○	●	○	Beta
F09	**Kit check**	Combined track + OHE checklist before patrol (RDSO TI/MI/0060 6.1(3) OHE tools)	All items required	REQ-11	●	○	○	○	Beta
RECORD & SYNC									
F10	**Offline store and sync**	Room + WorkManager (network constraint, exponential backoff)	No data loss on airplane-mode test	REQ-02	●	●	●	○	Beta
F11	**Combined patrol diary**	Auto-compiled in two parts: track (CAMTECH 2021 B17 fields) and OHE (foot patrol diary, Proforma 03-2 Category A/B)	Splits and routes correctly (API test)	PR-07	●	○	○	○	Beta
F14	**Audit trail**	Append-only SHA-256 hash chain; /audit/verify	Chain verified in tests	REQ-14	○	●	●	●	Beta
SUPERVISOR									
F12	**Routed supervisor review**	Inbox per department (SSE/P.Way, SSE/TRD); severity; confirm/reject; API R4	Patroller cannot review (API test)	REQ-05	○	○	○	●	Beta
F13	**Closure with evidence**	Closure evidence must pass quality gate; API R5	API test	REQ-09	○	○	○	●	Beta
LANGUAGE & ACCESS									
F16	**Language switch**	Per-app language (Android 13 system setting + in-app)	Hindi screens rendered in CI	—	●	○	○	○	Beta
F17	**Accessibility**	Large-text mode, ≥ 48 dp targets (HUD 60-88 dp), TalkBack labels, greyscale contrast	Stage V6	—	●	○	○	○	Beta